

Utilizing Artificial Intelligence for Categorizing Wildfire Origins through Satellite Image Analysis

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Abstract:

As climate change continues to create warmer and drier climates, wildfires are becoming more dangerous and more prevalent. One valuable area to study when investigating wildfire risk mitigation is the ignition cause of fires. Most fires are caused either by humans or lightning. It is easier and more productive to focus on preventing the more controllable human fires. In this research, a convolutional neural network was created to predict ignition causes from satellite imagery. The model was trained on two publicly available data sources. One contained satellite images from Quebec of areas that experienced wildfires before the wildfire occurred, and the other contained information on every sizable fire across Canada. These two data sources were combined to create a dataset of about 22,000 images which the model was trained and tested on. The final model was relatively accurate at differentiating between ignition causes with an accuracy of 73%. With more data, this accuracy could potentially be improved upon and applied to real-world wildfire risk management.

1. Introduction

Global warming has led to increasingly warm and arid conditions where wildfires thrive_[1]. These hot and dry climates are the perfect breeding grounds for fires to ignite and spread_[2]. As a result, wildfire risk is currently very high. The ignition of wildfires is key to understanding this risk. The United States Department of Agriculture defines three categories of ignition type: natural, accidental, and arson. Natural fires are largely caused by lightning, and therefore the risk from this ignition type is difficult to mitigate. However, accidental and arson fires are a result of human activity. Therefore, it is more feasible and productive to focus on reducing fire risk in areas susceptible to human-caused, rather than natural-caused wildfires_[3]. The goal of this project is to develop an AI model that can determine which type of ignition an area is most susceptible to from a satellite image.

2. Background

Some prior research exists in predicting fire ignition types. One study from Bountzouklis, Fox, and Bernardino in April 2023 used AI to analyze land cover, topographic, anthropogenic, and spatiotemporal features to differentiate between ignition sources_[4]. This study was able to classify natural fires with an F1-score of 0.87, accidental fires with an F1-score of 0.74, and arson fires with an F1-score of 0.64. Additionally, through the use of XAI methods, the study found that the two most important factors for classification were the spatiotemporal and topographic features.

Another study from the University of Alberta investigated the differences between natural and human fires_[5]. This research looked at wildfires that occurred in British Columbia and used machine learning to identify high-density clusters of human and lightning-caused wildfires. Intuitively, human-caused fires more often occurred near population centers. This study is not alone in this finding. Another paper from South Dakota State University, which studied fires in the United States rather than Canada, found that human-caused fires tended to be concentrated in high-density road areas, while lightning-caused fires were found more prevalent in low-density road areas_[6]. Less intuitively, the University of Alberta study found ignition types showed significant differences in their fuel. Human-ignition clusters were more often found within leafless aspen, ponderosa pine, and Douglas fir trees, while lightning-ignition clusters showed no significant trend to any fuel source.

3. Methodology

3.1 Data Collection

Two datasets were used for this project. They are separate Kaggle data sources, but they both originate from the same government-run Canadian National Fire Database (CNFDB)^[7]. This database “is a collection of forest fire data from various sources” and contains various information on individual wildfires that have occurred across Canada. The first Kaggle dataset from this source is *Wildfires in Canada (1950-2021)*^[8]. It is a CSV file containing numeric data from the CNFDB. Each fire is a row in the CSV and contains information on the fire ID, source agency, longitude, latitude, date reported, size, cause, protection zone, and ecological zone where the fire occurred. The other Kaggle dataset is *Wildfire Prediction Dataset (Satellite Images)*^[9] which contains JPG images of areas that previously experienced wildfires before the incident occurred. The coordinates for these areas are taken from the CNFDB and the images are titled by their coordinates.

All of these images are within Quebec, and only 22,710 exist compared to the nearly 500,000 fires in the entire CNFDB CSV file. By creating a coordinate matching program the longitude and latitude values in the CSV were matched with the image titles, assigning a fire type to each image. In some cases, multiple fires in the CSV database contained the coordinates in the image title. Because of this uncertainty, these images were discarded from the dataset, leaving 22,514 for use. In this set, approximately 70% of images are of areas where a human fire occurred and 30% are of areas where a lightning fire occurred. Below, a map of the location of wildfires in the data as well as example images are shown.

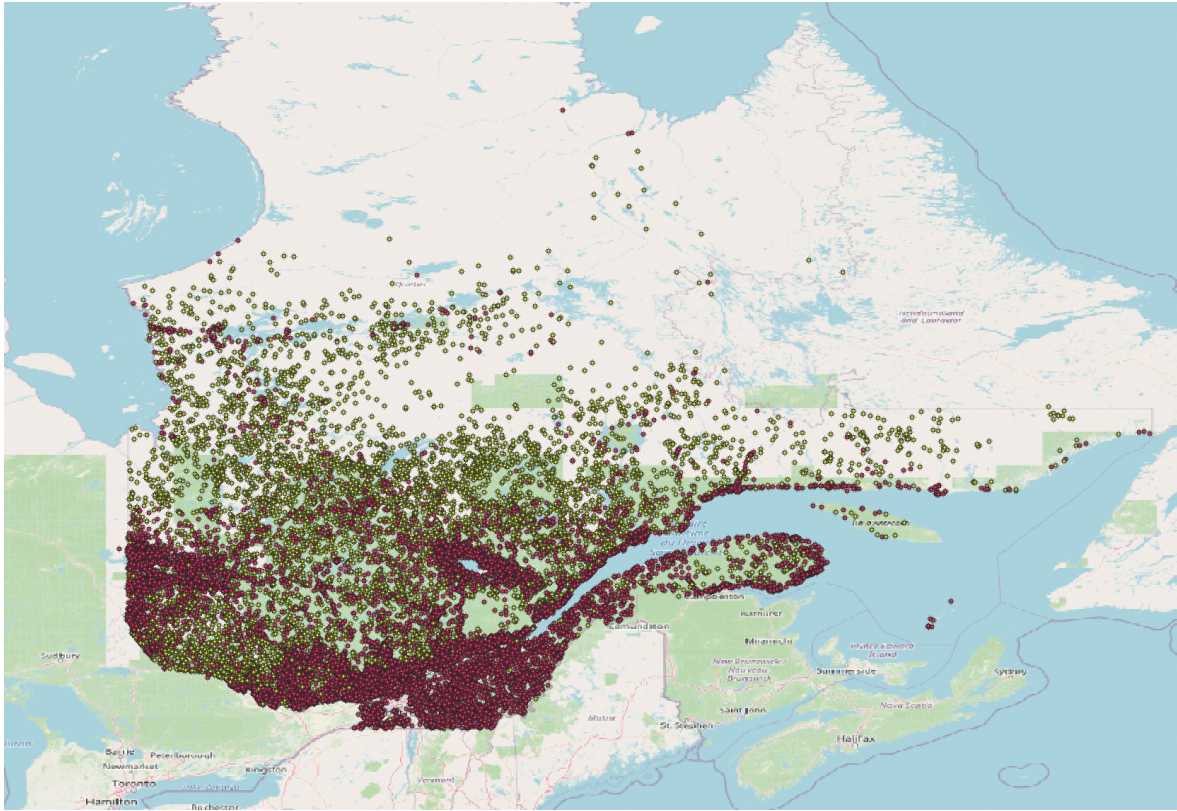


Figure 1: Plot of Satellite Image Locations

In this figure, red points signify human-fire locations and yellow points signify lightning-fire locations. It is evident that there are more human fires towards the southern border and more lightning-fires towards the north. This is likely because the population density is much greater near the American border than far north, so it is unsurprising there are more human-caused fires where there are more people.



Figure 2: Example Human Fire Satellite Image



Figure 3: Example Lightning Fire Satellite Image

In many of the human-caused fire locations, urban features, such as roads and houses can be identified. These are often absent in the lightning-caused fire locations and are replaced with dense, rural forested areas. These are not absolute rules but are general trends in the images.

3.2 AI Models

This research used Tensorflow to create a convolutional neural network (CNN). CNNs are primarily used for finding patterns in image data, so they are a good choice for this classification task. The first model created had the following architecture, which was chosen to create a standard baseline model that will be altered throughout trials:

```
model = tf.keras.Sequential(  
    [  
        tf.keras.Input(shape=(350, 350, 3)),  
        layers.Conv2D(32, (3, 3), activation = 'relu'),  
        layers.MaxPooling2D((2, 2)),  
        layers.Conv2D(64, (3, 3), activation = 'relu'),  
        layers.MaxPooling2D((2, 2)),  
        layers.Flatten(),  
        layers.Dense(64, activation = 'relu'),  
        layers.Dense(1, activation = 'sigmoid')  
    ]  
)
```

This architecture adheres fairly well to common practices and saw steady increases in training accuracy up to 99%. However, the testing accuracy remained at 66%. Because more than two-thirds of the images are of human fires, this is no better than guessing. The massive increase in training accuracy with poor testing accuracy indicated overfitting. Thus, dropout layers were introduced.

```
model = tf.keras.Sequential(  
    [  

```

```

tf.keras.Input(shape=(350, 350, 3)),
layers.Conv2D(32, (3, 3), activation = 'relu'),
layers.Dropout(0.2)
layers.MaxPooling2D((2, 2)),
layers.Conv2D(64, (3, 3), activation = 'relu'),
layers.Dropout(0.2)
layers.MaxPooling2D((2, 2)),
layers.Flatten(),
layers.Dense(64, activation = 'relu'),
layers.Dropout(0.5)
layers.Dense(1, activation = 'sigmoid')
]
)

```

This did reduce the training accuracy as expected but did not yield any increase in the testing accuracy. Looking to minimize potential issues with class imbalance, the dataset was balanced. Excess human fires were randomly removed from the data in order to achieve 50% human fires and 50% lightning fires. The same architecture was used. This model achieved a 62% testing accuracy, which is much improved from the guessing of the original models. Although the 66% accuracy of the previous model is a higher number, this 62% is more accurate because it was tested on a 50/50 data set rather than a 70/30 data set. Therefore, the 62% represents a greater improvement from random guessing. However, this model still showed overfitting, with a training accuracy of 94%. Tuning the dropout values failed to resolve the overfitting, so batch normalization layers were tested in an attempt to combat the issue. The model had the following architecture.

```

model = tf.keras.Sequential(
    [
        tf.keras.Input(shape=(350, 350, 3)),
        layers.Conv2D(32, (3, 3), activation = 'relu'),
        layers.BatchNormalization(),
        layers.MaxPooling2D((2, 2)),
        layers.Conv2D(64, (3, 3), activation = 'relu'),
        layers.BatchNormalization(),
        layers.MaxPooling2D((2, 2)),
        layers.Flatten(),
        layers.Dense(64, activation = 'relu'),
        layers.Dense(1, activation = 'sigmoid')
    ]
)

```

This model produced the best results yet, with a testing accuracy of 73%. Some overfitting still occurs, but the testing accuracy reached its greatest height yet. The consistent overfitting throughout the models suggests that there may not be enough unique training data for the model. In order to address this, data augmentation methods such as rotations and reflections were attempted. However, RAM issues prevented the processing of large new amounts of data.

3.3 Flow Chart and Brief Descriptions

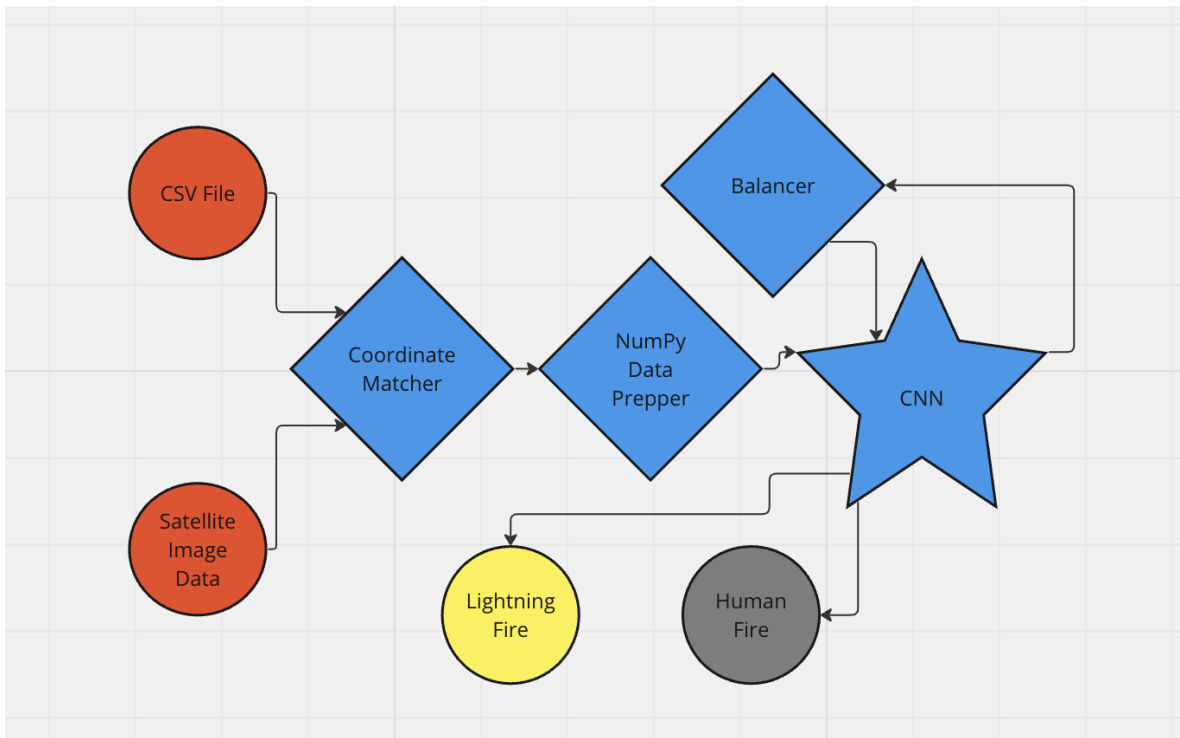


Figure 4: Flow Chart

- CSV File: Contains longitude and latitude values that match the image titles as well as the cause of each fire
- Satellite Image Data: Images titled with longitude and latitude which the AI model was trained on
- Coordinate Matcher: Matches the coordinates of the satellite images to the CSV file to assign a cause label to each fire
- NumPy Data Prepper: Formats images properly in NumPy and prepares for use in the CNN
- Balancer: Removes excess human fire images to create a 50% human 50% lightning dataset
- CNN: Trained to classify satellite images into human or lightning fire areas

4. Results

The best model could distinguish between human fire areas and lightning fire areas with a training accuracy of 92% and a testing accuracy of 73% when tested on an equal distribution of data. This was achieved with 15 epochs of training, a batch size of 128, and the final model architecture shown in the methodology section. The training and validation accuracy curves are shown below.

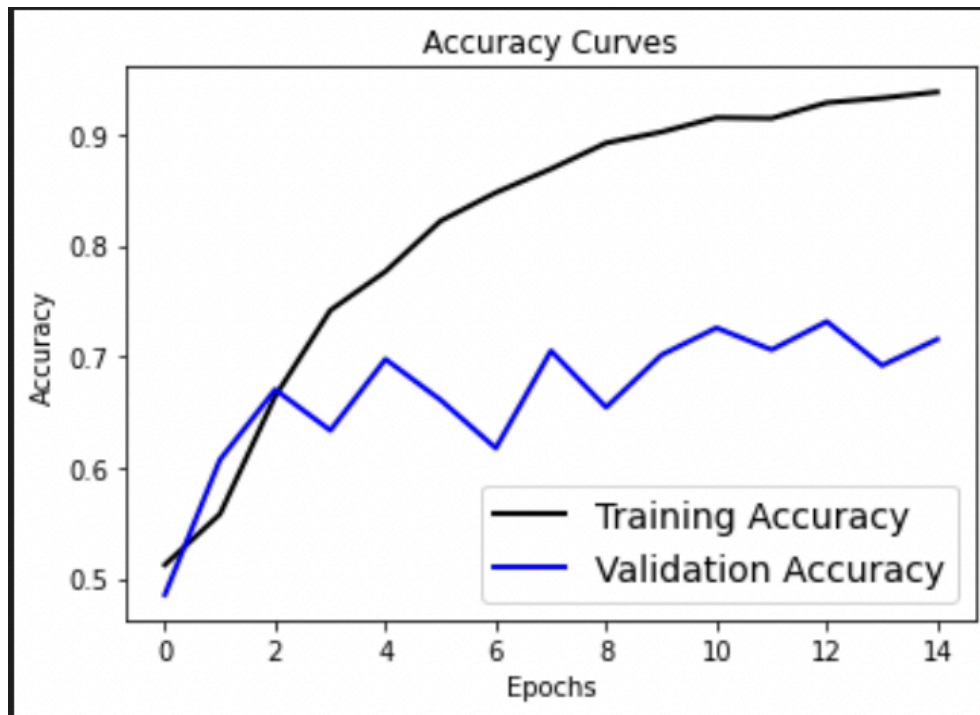


Figure 5: Training and Validation Curves

5. Discussion

In this research, a reasonably accurate AI model to distinguish between human and lighting wildfires was created. When compared with the most similar paper^[4], *Predicting wildfire ignition causes in Southern France using eXplainable Artificial Intelligence (XAI) methods*, the model's accuracy is slightly lower, with testing accuracy of 73% compared with F1-scores of 0.87, 0.74, 0.65, .64, and .55 for the five fire types the study attempted to classify and an overall accuracy of 69.8%. Of course, this study classified fires into five groups rather than two, so this similar accuracy is more powerful.

However, this higher accuracy was achieved with vastly different data. This research used solely satellite imagery taken before the fire, while the research focused on Southern France used large amounts of numeric data depicting the land cover and topographic, anthropogenic, and spatiotemporal features of the area. Importantly, their model was trained on data that could only be collected after the fire occurred, such as fire size and season. This places their model as a tool best suited for classifying previously unknown fire causes, rather than forecasting. Because the model created in this research only uses data taken before a fire occurs, it can more effectively be applied to forecasting. This difference in required data is the biggest contribution this study makes.

One of the largest limiting factors in this project was the amount of data. The model was trained on approximately 16,000 images. With an increase in images or through data augmentation techniques, accuracy may be improvable. However, it is likely that an extremely high accuracy, for example, 95%, may simply be impossible to achieve with any amount of satellite imagery. An image of an area can only tell so much, and there may be some cap to

possible accuracy. Still, it is unlikely that the model created in this project reached this cap due to the limited data available.

6. Conclusion

The USDA has stated areas of high human fire risk are easier to address than areas of high lightning fire risk. Therefore, it is important to be able to tell if an area is more at risk of a human or lightning fire so risk mitigation institutions can better allocate resources to places that will benefit. The model created could be an important tool in this risk analysis. Although the accuracy may at this point be slightly too low to use effectively, it could be improved in further research to serve this goal. Compared to other papers that attempt to classify fire ignition causes, this model excels due to both the ease of collecting data and the ability to collect this data before a fire occurs. Overall, this research was successful, as a relatively accurate model could be created from satellite imagery alone.

7. References

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